

NKS

CONTACT

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EDUCATION

B.Tech, Computer Science
Maharaja Agrasen Institute of Technology
2019 – 2023 • CGPA 8.4/10

Senior Secondary (Non-Medical)
GSS School, Baddi, HP
2018 – 2019 • 63%

Secondary (X)
Nav Jyoti Century School, CBSE
2016 – 2017 • CGPA 8.6/10

SKILLS

- Unity Engine (Expert)
- C# / .NET, C++, Python
- VR/AR/MR — Meta Quest, HTC Vive
- Unity Netcode, Photon Networking
- REST APIs, Sockets, PlayFab
- BLE, NFC, OpenCV, QR Tracking
- Maya, Blender, Photoshop
- Git, Gradle, WebGL Deployment
- ElevenLabs Voice AI
- Performance Optimization (90+ FPS)

LANGUAGES

English • Hindi • Punjabi

CERTIFICATIONS

Google IT Support — Internshala
Full-Stack Web Development — Internshala

CORE STRENGTHS

- Hardware–Software Integration
- Cross-Platform Deployment (VR/AR/Desktop/Mobile)
- Real-Time Systems & Simulation
- Rapid Prototyping

NIKHIL KUMAR SINGH

UNITY GAME & XR (VR / AR / MR) DEVELOPER

PROFILE

Unity developer with 2.9+ years of professional experience building immersive VR/AR/MR applications, hardware-integrated simulations, and multiplayer systems. Delivered 10+ production projects for international clients, including the NHS (UK) and the UK Army. Skilled in custom hardware integration (BLE, sensors, cameras, NFC, drones), computer vision with OpenCV, real-time networking (Netcode, Photon), and cross-platform deployment across VR, AR, mobile, and desktop.

HARDWARE & SYSTEMS INTEGRATION

Cross-project highlights connecting real-world hardware and sensors to Unity, XR headsets, and desktop applications:

- CBRN Training Simulation (MR):** built a desktop-to-Oculus command pipeline where operator commands from a desktop build execute as real-time in-headset actions within a Mixed Reality CBRN training scenario.
- Head-Tracking Camera Control:** developed a head-tracking feature where lateral head movement (left/right), captured via camera input, directly drives vehicle movement in a desktop simulation.
- NFC-Based Object Recognition:** engineered an NFC tag-reader system for 3D-printed models — scanning a tagged model against a reader auto-displays its full information set.
- Drone Simulator:** built a hardware-controlled drone simulator giving full desktop flight control (throttle, pitch, yaw, roll) synced with physical drone hardware.
- OpenCV QR-Based Interaction System:** implemented multi-QR detection mounted on a physical prop gun; scanning a QR code through the Oculus camera triggers a real-time in-headset firing action.
- ElevenLabs Voice Integration:** integrated multiple AI-generated voice profiles via ElevenLabs across simulation and training applications for dynamic narration.
- Wearable Vitals Streaming:** built a Wear OS app streaming live vitals (SpO2, BPM, steps) from a Galaxy Watch to desktop; reverse-engineered third-party pulse oximeters to stream vitals via BLE to desktop and Oculus.
- Physical Cycle → Mobile Game Sync:** extended BLE cycling-hardware integration to mobile, syncing real-world pedaling with in-game movement across multiple routes.

EXPERIENCE

Unity Game Developer / *BigVision, Mohali*

Sep 2024 – Present

- Developed MR DEMS for the UK Army — a mixed-reality mine detection simulation with controller integration and 3D data visualization.
- Built a CBRN Mixed Reality training simulation with a desktop command pipeline driving real-time in-headset (Oculus) actions.
- Created NHS Desktop, a hospital simulation with VR support, server-based asset management, and 360° environments for medical training.
- Delivered NHS Hand Washing app implementing WHO hygiene protocols through interactive PPE training modules.

- Client-Facing Project Delivery
- Team Collaboration

AT A GLANCE

2.9+ Years Professional Experience

10+ Projects Delivered

Clients across USA & UK

NHS (UK) & UK Army Projects

- Built VR Game Book, a Sikhism-themed educational VR experience with mini-games and historical location exploration.
- Shipped five AR book/marker-based experiences (Baishakhi, Chaar Sahibzade, House AR, Vedic Book) triggered by printed pages.
- Delivered the hardware-integrated systems detailed in Hardware & Systems Integration above, including head-tracking, NFC, drone, OpenCV QR, voice, and BLE vitals pipelines.

Unity Game Developer / *Blocktech Brew, Chandigarh* Mar 2023 – Aug 2024

- Engineered the NYC Pedal Adventure trilogy, syncing external Bluetooth cycling hardware with Unity across 3 environments.
- Implemented real-time BLE connectivity with dynamic difficulty adjustment based on physical rider input.
- Extended pedal-cycle hardware integration to a mobile build, syncing real-world pedaling with in-game movement.
- Developed AR DINO, an educational AR game with interactive dinosaur scanning and live data display.
- Built Art Gallery and Virtual Port Metaverse experiences, including an API-driven CMS and logistics simulation.
- Designed a Mahabharat-themed racing game with cultural storytelling and complex AI behaviors.
- Implemented multiplayer networking with Unity Netcode during the internship period.

KEY PROJECTS

- VR Game Book, MR DEMS, CBRN MR Training Simulation, NHS Hospital Simulation Suite
- AR Book Series (5 titles), NYC Pedal Adventure (Desktop + Mobile), AR DINO
- OpenCV QR-Based Firing System, Wearable & Oximeter Vitals Streaming (BLE)

ACCOMPLISHMENTS

- Delivered 10+ projects for international clients across the USA and UK, including the NHS and the UK Army.